

CCNA 200-301

CHAPTER 29

Securing Routers

Router Access Security: Console, Local Users, Enable Secret, Telnet and Management IP

Source: Eng. Walid Saad Eldin video - practical Cisco router security configuration.

Scope: This chapter follows the provided video content and command sequence.

Chapter Objectives

By the end of this chapter, you should be able to identify the main management access methods, secure console access, create a local user with secret, protect Privileged EXEC mode with enable secret, configure basic Telnet access, assign a management IP address, and save the configuration.

1. Introduction

The video explains the basic hardening of Cisco routers against unauthorized administrative access. The focus is on the three management paths discussed in the lesson: Console, Telnet, and the mention of SSH as a more secure alternative for a later lesson.

The same configuration principles demonstrated here apply to both routers and switches.

Core access model

Access method	What it means	Main command
Console	Direct physical access to the device.	<code>line console 0</code>
Telnet	Remote access over the network; the video notes that it is not secure because it uses plain text.	<code>line vty 0 4</code>
SSH	Mentioned as the more secure remote alternative; setup is not covered in this video.	Later lesson

Key idea: securing a router means protecting each management path and protecting the transition into Privileged EXEC mode.

2. Securing Console Access

The console port provides direct physical access. The video demonstrates protecting it with a line password and then enabling the login prompt.

Configuration

```
configure terminal
line console 0
password your_password
login
```

What each command does

Command	Purpose
<code>configure terminal</code>	Enters Global Configuration Mode.
<code>line console 0</code>	Enters the Console line configuration context.
<code>password [password]</code>	Defines the line password.
<code>login</code>	Tells the line to require the configured password during login.

Remember: the password is defined with `password`, but `login` is what makes the line ask for it.

3. Local User Accounts and Secret

The lesson then moves from a single shared line password to individual local credentials. The recommended command shown in the video is `username ... secret ....`

```
username admin secret MyPassword
line console 0
login local
```

Why this matters

Using `login local` makes the Console authenticate against the router's local user database instead of using the single line password method.

Command	What it does
<code>username [name] secret [password]</code>	Creates a local user and stores the credential using the router's secret mechanism.
<code>line console 0</code>	Selects the Console line.
<code>login local</code>	Uses the local user database for authentication on that line.

4. Securing Enable Mode

Console authentication controls access to the device through the Console line, but the router also needs protection at the Privileged EXEC level. The video uses `enable secret` for this purpose.

```
enable secret MyPassword
```

When the administrator enters:

```
enable
```

the router requests the configured enable secret before allowing entry to Privileged EXEC mode.

Important distinction

Console security and Enable security protect different stages: Console credentials protect access to the line, while `enable secret` protects the transition into the privileged mode used for higher-level administration.

5. Configuring Telnet

Telnet allows a network administrator to access the router remotely. The video demonstrates the VTY configuration and explicitly warns that Telnet is insecure because it sends data in plain text.

```
line vty 0 4
password MyPassword
login
```

Command breakdown

Command	Purpose
<code>line vty 0 4</code>	Enters the VTY line configuration used for remote access.
<code>password [password]</code>	Sets the VTY line password.
<code>login</code>	Requires the configured VTY password during login.

Security warning from the video: Telnet is not secure because credentials and session data are transmitted in plain text. SSH is presented as the more secure alternative, but its configuration is outside this chapter.

6. IP Addressing for Management

For remote management to work, the device needs an IP address on a reachable interface. The video demonstrates assigning an IP address and bringing the interface up.

```
interface gigabitEthernet 0/0/0
ip address [ip_address] [subnet_mask]
no shutdown
```

Command	Purpose
<code>interface gigabitEthernet 0/0/0</code>	Selects the GigabitEthernet interface used for management in the lab.
<code>ip address [ip] [mask]</code>	Assigns the IPv4 address and subnet mask.
<code>no shutdown</code>	Enables the interface.

7. Save the Configuration

```
do write  
or  
copy running-config startup-config
```

Both commands are used in the video to save the current configuration so the configuration is retained after a reload.

8. Complete Configuration Flow

A compact way to remember the lesson is to think of security as several layers of access control.

```

Console
-> line console 0
-> password / login
-> or username ... secret ... + login local

Privileged EXEC
-> enable secret

Remote access
-> line vty 0 4
-> password / login

Management interface
-> ip address
-> no shutdown

Save
-> copy running-config startup-config

```

9. Quick Command Reference

Task	Command	Effect
Enter global config	configure terminal	Move to configuration mode.
Secure Console	line console 0 password ... login	Protect Console with a line password.
Local user	username ... secret ...	Create a local authenticated user.
Use local users	login local	Use the local user database on the line.
Protect enable	enable secret ...	Protect Privileged EXEC mode.
Configure VTY	line vty 0 4	Enter remote VTY line configuration.
Interface IP	ip address ...	Assign IPv4 address and mask.
Enable interface	no shutdown	Bring interface up.
Save config	copy running-config startup-config	Save current config for next reload.

10. Exam Notes

- Console = direct physical management access.
- Telnet = remote management access and is sent in plain text according to the video.
- line console 0 = Console line configuration.
- line vty 0 4 = VTY lines used for remote access.
- login = use the line password.
- login local = use the local username database.

- `username ... secret ...` = create a local user using the secret mechanism.
- `enable secret` = protect Privileged EXEC mode.
- `no shutdown` = enable an interface.
- `copy running-config startup-config` = save the configuration.

Note: SSH is mentioned as the more secure remote alternative, but the actual SSH configuration is not part of this video.

11. Final Revision Sheet

```
SECURE THE ROUTER
=====
```

1) Console

```
line console 0
password <password>
login
```

2) Local authentication

```
username <name> secret <password>
line console 0
login local
```

3) Privileged mode

```
enable secret <password>
```

4) Remote access

```
line vty 0 4
password <password>
login
```

5) Management interface

```
interface gigabitEthernet 0/0/0
ip address <ip> <mask>
no shutdown
```

6) Save

```
copy running-config startup-config
```